

AWI–ESM3 v3.5 PI-control tuning campaign — summary

Forty rounds, from a boreal forest that vanished to a production line that reaches year 2100,
cools while gaining energy at the top, and still has no forest

OpenIFS 48r1 / FESOM2 / LPJ-GUESS / OASIS / XIOS, TCO95L91 / CORE3

supersedes the 2026–08–24 summary

2026–09–26

Abstract

The short reading of a campaign whose full record runs to 148 pages (`report.tex`). Three objectives throughout: the **global radiative imbalance**, the **Southern Ocean**, and the **NH high-latitude cold bias / boreal forest**. §4 is the whole campaign in the order it happened, rounds 1 to 40; the sections after it are the state now.

The founding finding still stands. Every lever that lowers the TOA imbalance does it by cooling the model. The 29 early runs lie on a line, not in a cloud: about 0.6 K of global cold per W m^{-2} of TOA. Underneath that line sits a structural floor that no atmospheric retune reaches — a boreal forest that LPJ-GUESS does not deliver, worth -3.1 K of Siberian DJF on a clean with/without pair.

The campaign has now left lever-screening behind and produced a single 160-year pre-industrial line, **PICAL_ccnice** (1940–2100), on the v3.5 defaults. It is not converged, and the drift is cold: $0.214 \pm 0.025\text{ K/decade}$ globally over 2060–2099, 0.511 ± 0.118 at 60–90 N, with March NH ice extent growing.

Positive TOA does not mean warming here. Net TOA over 2080–2099 is $+0.167\text{ W m}^{-2}$ while the surface cools. Total ocean heat content is flat ($+0.04\text{ ZJ/decade}$): the upper 700 m loses to below 700 m at 0.32 W m^{-2} equivalent, almost exactly balanced. **The cold drift is an ocean redistribution, not an energy leak**, and nothing has yet been aimed at it.

The Southern Ocean has resisted the whole campaign. It generates the imbalance rather than responding to it ($+0.383$ and $+0.353\text{ W m}^{-2}$ from the two southern bands), 84% of its cloud error is already present in AMIP over a correct ocean, and five more levers were excluded in round 37. It sits at 62% of observed winter ice volume and the ACC at 98.9 Sv against 134 ± 11.2 observed.

The boreal forest is gone. BNS 0.001 and 5% tree cover in the Siberian box at 2099, with grass beating tree on 84.9% of cells, every establishment gate open. This is now a practical blocker: the LPJ-GUESS canopy work cannot be evaluated in a configuration with no trees.

1 Scorecard against the three objectives

objective	at the start	what the campaign did to it	where it stands
Radiative imbalance	Coupled net TOA +1.44 to +1.73 W m ⁻² , and every lever that lowered it cooled the model 0.6 K per W m ⁻² .	Measured that $\approx 45\%$ is atmospheric (AMIP +0.67); re-based the target to 0; found that decade-4 TOA follows mid-depth heat burial ($r = +0.72$) and that GM is the lever that reaches it.	Partly understood, not fixed. +0.167 W m ⁻² over 2080–99 — but the surface cools, so the number no longer means what it did (§5).
Southern Ocean	Largest single regional term, +7.46 W m ⁻² period-clean, and a winter over-ice warm bias.	Established it is cloud <i>amount</i> at 60–45 S and <i>opacity</i> at 90–60 S; that 84% is inherited from AMIP; traced and priced the marine CCN path; excluded nine levers in total.	Open, and now two problems with a third constraint. 62% of observed winter volume, ACC 30% weak, and the GM setting that helped the CDW costs 16.7 Sv (§8).
NH cold bias / boreal forest	Siberian JJA –2.58 K against ERA5; cvh 0.63 → 0.02 when LPJ-GUESS is switched on.	Split it into a universal part, an orographic part and a boreal-specific part; recovered ≈ 1.0 K of the last with G4/K1 and another 0.77 with the stable-BL lever; closed land albedo and surface exchange as explanations.	Atmospheric half improved, vegetation half worse. Best coupled all-land bias –2.79 → –2.11 K, but the forest is effectively gone and the Arctic over-grows ice (§10).

2 Where the model stands

PICAL_ccnice, 2090–2099 unless stated. Observations as noted.

quantity	model	observed	source
net TOA (2080–99)	+0.167 W m ⁻²	0 for a control	—
net surface (2080–99)	+0.357 W m ⁻²	0 for a control	—
T2m global trend (2060–99)	–0.214 K/dec	0 for a control	—
NH April ice volume	34.56	31.19	GIOMAS 1989–99
NH September ice volume	13.97	11.96	GIOMAS 1989–99
SH September ice volume	12.33	19.89	GIOMAS 1989–99
SH February ice volume	0.73	2.07	GIOMAS 1989–99
Drake Passage transport	98.9 Sv	134 ± 11.2	Cunningham 2003
		173.3 ± 10.7	Donohue 2016
cavity basal melt	1062 Gt yr ⁻¹	~1100	Adusumilli 2020
		~1500	Rignot 2013
CMPI (global)	0.903	lower is better	—
Siberian tree cover (FPC)	0.050	0.415	offline CRUNCEP

Volumes in 10³ km³. The NH is expected *above* a satellite-era record for a pre-industrial run; the SH is close to like-for-like because GIOMAS has no significant Antarctic trend across it.

3 What v3.5 carries, and where each number came from

The tuning now lives in `configs/setup/awiesm3/awiesm3.yaml` rather than in runscripts, so a fresh install reproduces the line. Everything below is namelist-only except where noted, which is what makes it mergeable with EC-Earth.

parameter	value	why, and which round decided it
RCL_INPSEA	0.1	D2b, round 11. Reduced marine ice nuclei → brighter supercooled Southern Ocean cloud.

parameter	value	why, and which round decided it
RCL_INPPMIN	70000	Round 28 adopted 50000 (S4) and round 38 retired it: the benefit reverses sign coupled. 70000 follows <code>suc1dp.F90</code> 's own recommendation.
RSBLB	2.0	Rounds 31–32. The winter land lever; Arctic DJF screen $+0.614$ K coupled at net TOA -0.001 .
RCCNSEAICE	15.0	Round 38. Open-water weighting of the whitecap sea-salt source, which OpenIFS otherwise evaluates over sea ice. Gives the current line its name.
RVRSMIN(3,4)	1000	Round 14 (G4). A cap set by winter damage, not a saturation knee.
ECE_SNOW_SCF	3 (+9 coeffs)	Round 20 (P3/P5), fitted to Russian snow-course observations after the <code>tanh</code> form was falsified against 174 676 station soil records.
RVVEGALB	40 values	Round 17 (K1). Snow-free land albedo corrected against CERES; closed land albedo as an explanation.
cavity_gamma_scale	0.6	Round 37. 1190 Gt yr^{-1} against 1893 uncalibrated, 1100–1325 observed. FESOM PR #1064, merged.
K_GM_max / Redi_Kmax	2500 / 1000	Round 36 (16E). Lowest TOA plateau kept. §8 is the bill nobody scored.
mix_scheme	<code>cvmix_TKE</code> <code>+cvmix_IDEMIX</code>	Round 39, four-way 20-year comparison from one restart.
use_momix	false	Round 38; the production line's first branch tests it.
albsn / albsnm	0.80 / 0.65	Round 38, computed from the model's own melt seasons; FESOM ships 0.81/0.77, so these are deliberate. <code>albi/albim</code> stay at FESOM's 0.70/0.68 — darkening them was falsified, they are pond-free.

Deliberately unset: `RSNOWLIN2` stays at the code default 0.030. Raising it to 0.04 is the only lever in fifty arms that moves the tropical longwave deficit ($+2.74 \text{ W m}^{-2}$, 120–127 % of it) and it warms the Southern Ocean too — but it costs $+0.42 \text{ W m}^{-2}$ globally. It remains the campaign's most interesting unadopted parameter.

4 The campaign in the order it happened

One paragraph per phase, with the number that settled it. The full record of each is in `report.tex`, whose §1 carries the same order.

4.1 The trade-off, and the floor underneath it (2026–07, rounds 1–8)

Twenty-nine coupled runs, screened on net TOA against the ERA5 temperature bias, produced the campaign's founding picture: **the runs lie on a line, not in a cloud**. Every W m^{-2} of TOA improvement cost about 0.6 K of global cold, from 080a at (1.73, $+0.28$) to 10A at (0.57, -0.93); the coupling and ocean-mixing family cut the imbalance 30–57 % but pushed the planet 1.7–2.1 K too cold. Centred RMSD was 1.41–1.64 K across all 29 runs, a ± 8 % spread against a raw-RMSD spread of 64 %: **the runs differ by an offset, not by a pattern**. That identified a shared structural floor rather than a tuning deficiency, and sent the campaign looking for a warming lever instead of another retune.

4.2 The floor is LPJ-GUESS, on a clean pair (2026–07–14)

A purpose-built no-LPJG reference — same CORE3 ocean, same TCO95 atmosphere, matched branches, HT-ESSEL with prescribed satellite vegetation, 30 years — differs from the baseline in LPJ-GUESS alone. **Siberian high-vegetation cover `cvh` goes** $0.63 \rightarrow 0.02$ when LPJ-GUESS is switched on (pan-boreal $0.49 \rightarrow 0.07$), and Siberia is -3.1 K colder in DJF. The no-LPJG run also beats the baseline on global T2m RMSD (1.514 against 1.717 K) and on CMPI (0.738 against 0.745) at essentially unchanged imbalance, so the warming lever is real and nearly free. The cold effect is present in every month except high summer, which already ruled out snow-albedo masking as the whole mechanism: that is a sunlight effect and cannot explain an equally cold polar night.

4.3 The feedback-free AMIP reference (2026–07–27/28)

An AMIP testbed was built so the atmosphere could be scored with the vegetation–albedo feedback open but the coupled runaway severed. **(i) About 45 % of the imbalance is atmospheric:** AMIP net TOA is $+0.67 \text{ W m}^{-2}$

against a coupled +1.44, and with SST prescribed that cannot be ocean heat uptake — the best branch at the time sat only 0.25 W m^{-2} above that floor, so the Part-I lever family was nearly exhausted. **(ii) The boreal summer cold bias is atmospheric too:** AMIP reproduces essentially all of the coupled Siberian JJA bias (-2.16 against -2.19 K) while its *annual* bias is only -0.89 against -3.0 coupled. Summer became an atmospheric problem and September–April stayed the coupled/vegetation one.

4.4 Rounds 10–12: a noise floor, a dead namelist, and the best boreal lever (2026–07–30→08–02)

Three infrastructure findings changed how everything afterwards was measured. The **4-year window was resolving nothing** in the boreal, so arms went to 46 years and a detection threshold became mandatory. The **period offset** was measured rather than estimated ($+0.42 \text{ K}$ for Siberian JJA, against a 1.12 K guess that had already been used to withdraw a conclusion). And **NCMIPFIXYR sat in a dead &NAMECECMIP6 block:** the AMIP runs were never at 1850 forcing, GHG ran 1869→1911, and Krakatoa is visible in 1883–86 in every arm. Deltas were unaffected, the absolute baseline was offset $\approx 0.15 \text{ W m}^{-2}$, and the coupled runs were correctly pinned. On the science side D2b (ice nuclei) was adopted for the Southern Ocean and F4 — boreal surface exchange — became the best lever to that point, with $G1 = F4 + D2b$ the best configuration.

4.5 Where the bias is not: the vertical structure (2026–08–02)

Before spending more levers, the column was decomposed. The Siberian cold bias is **confined below 850 hPa; the soil and skin are the least-biased parts of the column;** and **all 20 surface types are cold in every season**, so it is not a vegetation-tile problem. That decomposition eventually retired the surface-side lever family as a class, and it split the boreal budget into $\approx 0.7 \text{ K}$ universal, ≈ 0.3 orographic and $1.3\text{--}1.6$ boreal-specific — with G4 taking ≈ 0.95 of the last, and the remainder not reachable by a boreal-indexed lever at all.

4.6 Rounds 13–17: land albedo, and half of it is the snow tile (2026–08–03→05)

Half the Siberian shortwave deficit turned out not to be cloud. Of a $+0.0154$ land albedo excess against CERES, $+0.0074$ is the snow tile and $+0.0080$ the snow-free surface; every forest type is inside $+0.006$ once snow is masked per cell per month. The signature is **snow-melt timing** — accumulation is right, the melt is a month late and 30 % too slow, and May is a runaway. Round 14's G4 was adopted (Siberian JJA $+0.952 \text{ K}$, $t = 7.68$, SO SW RMSE $6.88 \rightarrow 4.80$) with RVRSMIN capped at 1000 by winter damage, and round 17 adopted $K1 = G4 +$ the snow-free albedo correction ($+1.036 \text{ K}$, global T2m RMSE $1.579 \rightarrow 1.524$, no Arctic energy penalty). Rounds 13 and 16 — snow cover fraction and skin conductivity — were both rejected on their own predictions.

And then land albedo was closed as an explanation. Removing 74 % of a directly measured albedo bias bought 0.089 K of a $\approx 0.7 \text{ K}$ target; a perfect fix buys $\approx 0.12 \text{ K}$, one sixth. That is the shape of most of this campaign: the diagnosis was right, the leverage was not there.

4.7 Rounds 18–21: the snow scheme, falsified on observations and refitted (2026–08–05→07)

The strongest summer lever built (I3, $+1.322 \text{ K}$ Siberian JJA, closing the surface SW deficit to CERES exactly) was *not* adopted, because it cost $+1.72 \text{ W m}^{-2}$ at $60\text{--}90 \text{ N}$. The sub-grid snow depletion scheme that HTESSEL lacks was implemented, validated offline, and it worked — melt-out 12 May against a satellite 11 May — and it cost -16.2 K of coupled DJF soil for $+1.2 \text{ K}$ of summer. The **tanh** functional form was then **falsified against observations rather than preference**: its range is open at 1, so complete cover is unrepresentable, while Russian snow courses report exactly 10/10 in 99.74 % of 14369 Siberian DJF surveys, and the scheme costs -20.2 K against 174676 station soil observations where the scheme *off* is right to $+0.8 \text{ K}$. The damage was distributional, not a matter of mean cover: DJF cover is identical to 0.002 across the arms while the soil differs by 23 K . P3, refitted to observations, was adopted; P5 followed. **Direct station data (RIHMI-WDC) entered the campaign here**, and the campaign stopped depending on ERA5 — an HTESSEL sibling — for any soil claim.

4.8 The forest fails by two routes (2026–08–05)

BNE/BINE are *gate-excluded*: 0.7 % of NE-Siberian cells clear `tcmin_est` = -30 and their LAI is exactly 0.000. BNS (larch) has *no* cold gate, clears its GDD gate on 84 % of cells, and is *outcompeted* — grass carries $5.8\times$

its LAI. Since NE Siberia is $\approx 80\%$ larch, **the winter-gate mechanism does not explain the region that matters**. This distinction has been rediscovered several times since and is why §10 states it explicitly.

Round 27 then measured the transfer function that the campaign’s founding premise assumed: offline 2000-year LPJ-GUESS equilibria give TREE 0.202 / BNS 0.034 under AMIP forcing, 0.261/0.058 at +1 K, 0.327/0.099 at +2 K, against 0.415/0.227 under observed forcing. Monotonic, not saturating, and +2 K recovers 59 % of the gap — **closing the cold bias should recover most of the forest**. A separate transfer test showed **forcing alone halves boreal tree cover**: same parameters, same 10 043 gridcells, `global.ins` byte-identical, Siberian TREEFPC 0.357 $\rightarrow$ 0.163 (-54%) when the spin-up forcing is swapped from CRUNCEP to what the model delivers.

4.9 Rounds 22–23: the energy target re-based, and the Southern Ocean promoted (2026–08–09)

Comparing a PI run against a 2000s CERES window conflates the climate difference with the bias. Measured period-clean, **the radiative error is $+1.23 \text{ W m}^{-2}$, about twice what the PI framing reported**. The tropical deficit largely dissolved (-2.51 PI-vs-CERES against -0.67 period-clean); the Southern Ocean got *larger* ($+7.46 \text{ W m}^{-2}$, $+0.735$ of the global mean) and became unambiguously the largest single regional term. Its deficit is **cloud amount, 65.5 % of it**, zonally uniform to 1.42 pp — and no lever in the campaign moves cloud area, the best managing $+0.63$ pp of a 6.43 pp gap. The reference question was settled too: CERES and MODIS put SO cloud area at 89.5 and 89.3 % against the model’s 83.1, while **ERA5 reproduces the model’s bias because ERA5 is IFS** and is therefore not an independent cloud reference. Meanwhile the global cold bias was located at the tropopause, peaking at 200 hPa over both poles, with the atmosphere cold *and* dry (-1.62 K and -10.1% over 700–300 hPa) in a pair that cancels in OLR.

4.10 Rounds 24–26: marine biogenic CCN, traced end to end and priced out (2026–08–09 $\rightarrow$ 10)

The one lever that reached the Southern Ocean in the right season. Droplet number there is **correct in the mean** ($N_d \approx 73 \text{ cm}^{-3}$ against 60–120 observed) but **inverted in phase**: the model is lowest in DJF because it is wind-driven, where observations peak in DJF from biogenic sulphate, and the CRE error is worst in exactly those months. A DMS emission climatology already shipped at TCO95 and nothing read it; it was wired through `ocp-tool`, defaulted off, and screened. At $S = 166$ it moves SO SW CRE -7.25 W m^{-2} in DJF, 67 % of the deficit in the worst season, from the bloom rather than a tuned constant. **It still cannot be adopted**: the tropical cost is 0.0129 W m^{-2} per unit S , so the tolerance binds at $S \approx 39$, below the published floor of 43, and scaling cannot separate the bands because tropical background CCN is *lower* than the Southern Ocean’s.

4.11 Rounds 27–28: a tuned stack nobody ran, and S4 (2026–08–10 $\rightarrow$ 11)

A jointly-tuned eight-parameter cloud stack for TCO95L91 was found sitting unused in the v3.4.2 DARS2 run-scripts, with no `fort.4` on disk setting any of it; it had been invisible because `preflight` looked in one place. Taken apart, it yielded `RSNOWLIN2`, still the only lever that moves the tropical longwave. The round also established that **Southern Ocean levers must be scored on net, not SW CRE** — longwave cancels a consistent 25–29 % across the 44-year archive. Then S4 (`RCL_INPPMIN` 70000 $\rightarrow$ 50000), one namelist number that had sat unevaluated for two days, came out as the best configuration the campaign had: SO surface bias $+3.66 \rightarrow -0.96$, global net TOA $+0.64 \rightarrow +0.25$. **It was the headline of the 2026–08–11 summary and is now retired** — its benefit reverses sign coupled (Siberian JJA $+1.194 \text{ K}$ AMIP against -0.365 K coupled). The habit of striking falsified claims through rather than deleting them dates from this episode.

4.12 Rounds 29–30: the land-surface repair, Raupach, and an inherited bias (2026–08–13 $\rightarrow$ 19)

Raupach canopy roughness is dead structurally, not statistically. Its winter mechanism needs a canopy, and the model’s Siberian trees are 2.89 m tall with 2 cm stems, giving $z_0 = 0.0132 \text{ m}$ against IFS’s 0.058 — it makes the winter surface $4.4\times$ smoother. **The lever cannot bootstrap the forest it requires**. The land-surface repair (107 new `fail()` calls classified, peat and LUH3 reconciled to one epoch) is a trade: $+0.308 \text{ K}$ of Siberian DJF soil against a JJA bias moved from -0.673 to -1.123 . Round 30 then produced the reframing that governs the Southern Ocean work since: **84 % of the coupled SO cloud error is already present in AMIP** over a prescribed, correct ocean. It is not a coupling problem. Round 30 also killed the “ -1.0 W m^{-2} structural leak”: it was the omitted snow-enthalpy term, and with it the column conserves to $0.02\text{--}0.04 \text{ W m}^{-2}$

and a TOA-zero run drifts -0.02 K/century . **Tuning net TOA to zero does stop the ocean drift** — which is what makes §5 surprising.

4.13 Rounds 31–32: the stable boundary layer, and the campaign’s best run (2026–08–21→23)

The decomposition said the winter land bias sits between the screen and 925 hPa: in the best arm the air-side inversion bias is $+2.81\text{ K}$ while the skin-side bias is $+0.11\text{ K}$. That excluded every surface-side lever by construction and pointed at stable mixing, so **&NAMVDFS** was created to expose it. **RSBLB 2.0** gives Arctic DJF screen $+0.614/+0.674\text{ K}$ coupled, and **it is free in energy terms coupled** — global net TOA $-0.001/-0.027$, both null, against AMIP’s $+0.175$: the tropical shortwave gain is absorbed as ocean warming and cancelled in the longwave. **Winter and summer moved together for the first time in the campaign.** 11R scored CMPI 0.7577 against 11E’s 0.8237, the best in the campaign. The mixing-length route (SB3) was falsified: its tropical SW cost per K of inversion is 3.43 against RSBLB’s 0.88, so the trade is a property of the scheme, not of the knob. Only about half the residual land bias is reachable by any mixing lever — of -2.23 K , ≈ 1.0 is excess inversion and ≈ 1.2 is a deep cold bias still present at 850 hPa.

4.14 Round 33: the Southern Ocean is two problems (2026–08–23)

It generates the imbalance rather than responding to it: 90–60 S and 60–45 S contribute $+0.383$ and $+0.353\text{ W m}^{-2}$, the two largest positive terms. The two bands need *opposite* corrections — at 90–60 S cover is right to 2 pp but each cloud is 8 % too dim, at 60–45 S there are 5 pp too few clouds each 9 % too bright — and every campaign lever raises opacity, which is why one band is now overcorrected while the other is not. Cloud phase flips sign at 60 S, and the INP family had been retired on a measurement taken in the one band where the model is too liquid. The ice melts from above in summer and fails to refreeze from below in autumn, with a $+0.9\text{ K}$ anomaly at exactly 100 m that autumn mixing entrains. **The ice is not too dark** (compact-ice albedo 0.760–0.778); it is too diffuse — no cell exceeds 90 % concentration in January–March, so the area-mean albedo falls to 0.49–0.51.

4.15 Rounds 34–36: where the heat goes (2026–09–06→12)

Round 34 found four defects in the coupled ice skin (§14). Round 35 asked where the heat lands: **below 300 m**, with the top 100 m gaining $+0.013\text{ W m}^{-2}$ while 300–2000 m gains $+1.388$; the tropical mixed layer is *not* too deep (43.6 against WOA18’s 51.6 m); the tropical deep warming is **heave, not diapycnal flux**; and heat crosses density surfaces at the mid- and high-latitude winter outcrops where the mixed layer is $+48.9$ to $+155.7\text{ m}$ too deep. The Southern Ocean *loses* heat rather than burying it, with 18.7 % of month-nodes mixing to the sea floor. Round 36 closed the loop: **decade-4 TOA follows mid-depth heat storage** across 20 runs ($r = +0.72$), **stronger GM is the lever that reaches it** (GM 1500 cuts first-decade 700–2000 m uptake by $\approx 30\%$), and 16E (K_GM_max 2500, Redi 1000) settles at the lowest TOA plateau the campaign kept, $+0.70\text{ W m}^{-2}$. The costs were stated at adoption: NH March ice $+1.3\text{ M km}^2$, global T2m $+0.21\text{ K}$, and an AMOC that GM 1500 strengthens and GM 2500 gives back. **Drake Passage was not among the costs scored (§8).**

4.16 Rounds 37–40: the production line (2026–09–13→25)

Round 37 spent five branches on the Southern Ocean winter bias — salt plumes, Fox-Kemper MLE, a deeper lead-closing floor, all three together, and a calibrated basal melt — and **excluded all of them**. It is a freshwater regime, not a mixing deficiency. Round 38 moved the v3.5 defaults into `awiesm3.yaml`, started the line, and computed the sea-ice albedos from the model’s own melt seasons rather than guessing: the four `ice_therm` albedos are *pond-free*, so darkening them to stand in for ponds double-counts, and the two snow albedos act in different hemispheres by measured surface state ($\partial\alpha/\partial\text{albsn}$ 0.908 in SH DJF against 0.149 in NH JJA). Round 39 scored four vertical mixing schemes and adopted `cvmix_TKE+cvmix_IDEMIX`, finding two real FESOM bugs on the way. Round 40 carried the line to 2100 and measured what it had become — which is the rest of this document.

5 The production line, and the cold drift

PICAL (1850–1929) → **PICAL_momixoff** (branched 1920, to 2049) → **PICAL_ccnice** (branched 1940, to 2100). Throughput on levante, 47 nodes, 10-year legs: about 77 SYPD under TKE+IDEMIX, 86 under

`cvmix_TKE` alone, 75 under KPP, with $\pm 15\%$ decade-to-decade scatter on identical configuration.

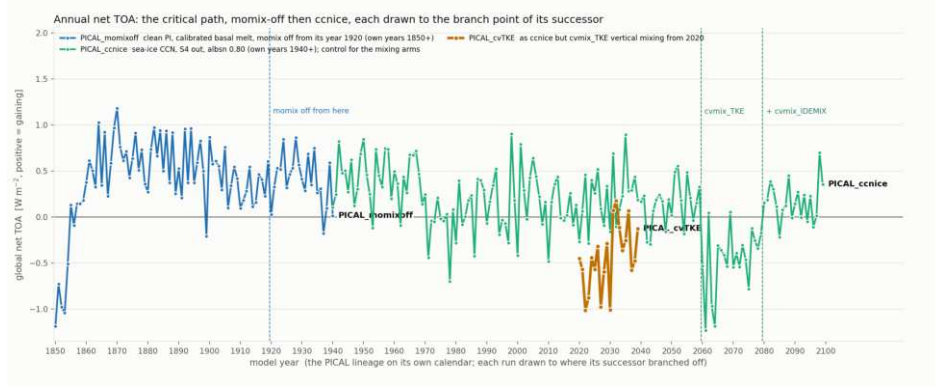


Figure 1: Annual net TOA along the critical path, each run drawn to the branch point of its successor. The two vertical marks are the mixing-scheme changes. `cvmix_TKE` at 2060 drives TOA negative; adding IDEMIX at 2080 brings it back. The run is never far from zero and never settles on it.

metric	2060–64	2095–99	trend/decade	95 % CI
T2m global	14.439	13.645	−0.214	0.025
T2m 60–90 N	−8.143	−9.842	−0.511	0.118
T2m 60–90 S	−15.650	−16.875	−0.238	0.104
NH extent, March	16.262	18.188	+0.522	0.103
net TOA	−0.775	+0.236	+0.272	0.079

Round 30 showed that tuning net TOA to zero *does* stop the ocean drift, so a run with positive TOA and a cooling surface needs explaining. It resolves in the ocean: total heat content is flat at $+0.04 \text{ ZJ/decade}$ over 2060–2099, while the bands are not (-17.8 , -34.4 , $+19.4$, $+32.7 \text{ ZJ/decade}$ for 0–100, 100–700, 700–2000 and $>2000 \text{ m}$). **The upper 700 m is draining into the deep ocean at 0.32 W m^{-2} equivalent.** Split by mixing era, the mechanism tracks the scheme:

window	upper 0–700	deep >700	total	net TOA	net surface
KPP 2040–2059	+0.061	+0.164	+0.225	+0.136	+0.318
<code>cvmix_TKE</code> 2060–2079	−0.488	+0.176	−0.312	−0.468	−0.251
TKE+IDEMIX 2080–2099	−0.201	+0.452	+0.251	+0.167	+0.357

All in W m^{-2} equivalent; the surface budget includes the snow-enthalpy term, about 0.9 W m^{-2} globally, without which the budget is biased warm. `cvmix_TKE` alone flipped the upper ocean hard negative and the whole system lost energy. Adding IDEMIX halved the upper-ocean loss and drove deep uptake up, which is what internal-wave-driven diapycnal mixing should do. But the upper ocean is *still* losing heat, and that is what keeps the surface cooling. **This is the top open item.**

6 Vertical mixing: the four-way comparison

Four schemes, 20 years each (2020–2039) from one restart, differing only in `mix_scheme`.

arm	scheme	SH Feb vol	SH Feb ext	cavity melt
PICAL_ccnice	KPP (control)	—	—	1255.6
PICAL_cvKPP	<code>cvmix_KPP</code>	—	—	1283.5
PICAL_cvTKE	<code>cvmix_TKE</code>	1.31	1.95	1174.5
PICAL_cvTKEIDEMIX	<code>cvmix_TKE</code> + <code>cvmix_IDEMIX</code>	1.62	2.26	1062.1

Observed SH Feb volume about 2.07, extent about 3.0. Both KPP variants lose Antarctic summer ice, both TKE variants grow it, and `cvmix_KPP` reproduces the KPP control closely enough to be the null. **Adopted**, with the cost stated at adoption: NH April volume 38.04 rising $+2.46/\text{decade}$ in the test arm, where TKE alone was flat at 35.17. On the production line the realised rate is $+1.145 \pm 0.968$, about half.

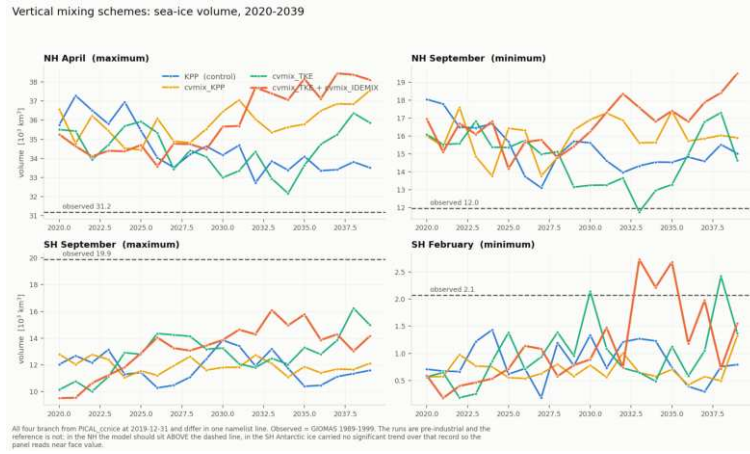


Figure 2: Hemispheric sea-ice volume across the four mixing arms, 2020–2039 from one restart. Both KPP variants lose Antarctic summer ice where both TKE variants grow it, and `cvmix_KPP` tracks the KPP control closely enough to serve as the null that shows the comparison is measuring the scheme rather than the branch.

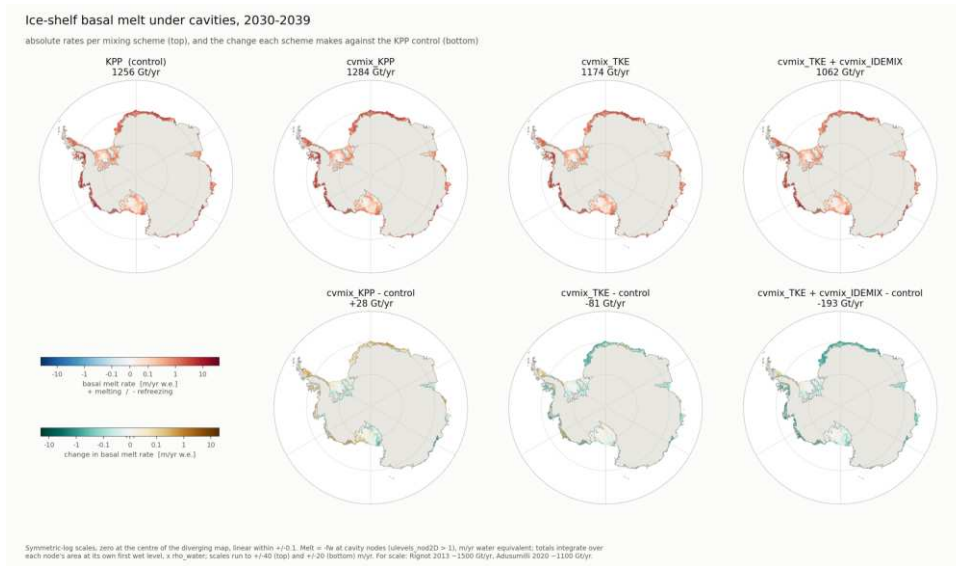


Figure 3: Ice-shelf basal melt under the cavities, 2030–2039, per mixing scheme (top) and the change each makes against the KPP control (bottom). Both TKE arms cool the cavities almost everywhere around the coast, with the largest reductions on the East Antarctic fringe and in the Amundsen/Bellingshausen sector; Ross and Filchner-Ronne interiors barely move. Symmetric-log scale, so refreezing keeps its own sign.

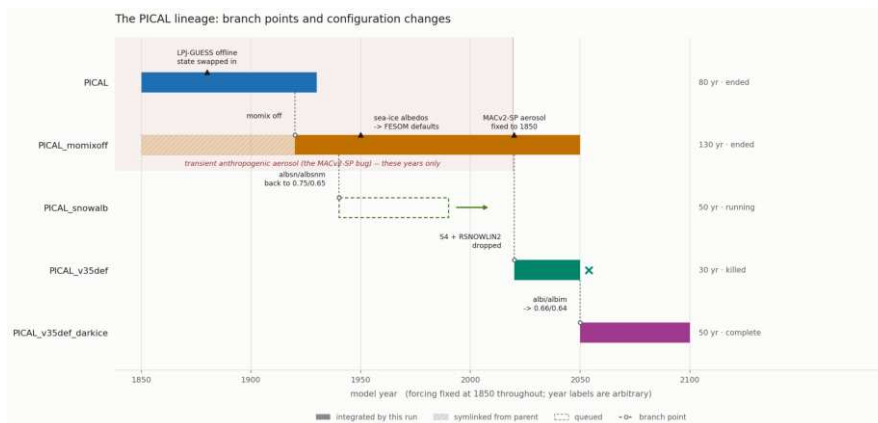


Figure 4: The PICAL lineage and every branch taken off it, on the model calendar. The production line is the spine; the arms hang off it at the year each was branched. The arms are cheap and the spine is expensive, which is the intended shape — but the spine has now run 160 years without converging.

7 The Southern Ocean now

Five more levers, all excluded (round 37). Four 20-year branches of **PI200** at 1560 against the PI200 control on identical years — **spp** (salt plumes), **mle** (Fox-Kemper), **h0** (lead-closing floor $0.5 \rightarrow 0.25$ m), **all3** — plus **cavg06** attacking the same loop from the ice shelves. None moves the winter over-ice bias in 20 years. The model sits in one of two freshwater regimes, salty-and-venting or fresh-with-warm-CDW, and these levers do not cross between them. The basal-melt calibration is worth keeping on its own terms (1-year screens gave $1038 / 1190 / 1402 \text{ Gt yr}^{-1}$ at $0.5 / 0.6 / 0.7$ against 1893 uncalibrated) and is upstream as FESOM PR #1064, but it is not a Southern Ocean fix. **SSS restoring stays excluded**; **surf_relax_s** is 0 and remains 0, even for Weddell convection.

The cloud shortwave deficit remains the other half. Across the whole coupled campaign no lever moved it until the mixing schemes: SW CRE at 45–65 S is -64.44 under KPP, -65.73 under **cvmix_TKE**, against CERES -68.02 W m^{-2} . **cvmix_TKE** closes about a third of the gap, more than any atmospheric lever achieved, and does it from the ocean side. That is worth recording: it is the one result suggesting the cloud error is not purely atmospheric — against round 30’s finding that 84 % of it is already present in AMIP.

8 Drake Passage, and the price of the GM tuning

Computed with tripyview’s model-accurate transect path, which resolves the 105 mesh edges the 66 W line crosses and integrates the flux FESOM’s own discretisation carries.

run (1580–1599, identical restarts)		GM	resolved	net
PI200	resolution scaling (before)		110.0	110.0
PI200_gmR1500	Rossby cutoff, K_{GM}^{max} 1500		103.2	103.2
PI200_gmR2500	Rossby cutoff, K_{GM}^{max} 2500		93.3	93.3

16.7 Sv, 15 %, monotonic in K_{GM}^{max} , far outside the ~ 1.5 Sv decadal scatter. The bolus term responds cleanly ($\pm 1.8 \rightarrow \pm 2.7 \rightarrow \pm 3.6$ Sv) but its *net* is zero in every arm: the eddy-induced circulation across the passage is a closed overturning cell. On the production line the transport declines slowly, 105.1 (2040–49) to 98.9 (2090–99), with no step at either mixing-scheme switch, so the ACC weakness is structural rather than a cvmix artefact.

GM 2500 was adopted in round 36 for the mid-depth heat uptake, with the NH ice and AMOC costs priced. Drake Passage was not scored at the time. Both constraints are now on the ledger and they point opposite ways.

9 Sea ice: volume is the state variable

Against GIOMAS 1989–99 the NH sits above observations in every month, which a pre-industrial run should, but by more than the epoch offset justifies in April; the SH is the observed cycle scaled down by about a third in every month, and that comparison is close to like-for-like because GIOMAS has no significant Antarctic trend across the window. Extent, scored against OSI-SAF, is reported beside volume throughout — with the caveat that the record on disk has no December, 286 months rather than 312, and a December value must not be synthesised. Extent is geometrically saturated in this configuration and has hidden a real Arctic runaway before: across the 1950 albedo step it barely moved while volume climbed from 24 to $48 \times 10^3 \text{ km}^3$. **Volume is the state variable that says where the pack is going**; extent is the one that gets compared to satellites. An ice-affecting lever is scored on volume first, and a null result in extent is not evidence that nothing happened.

10 The boreal forest, and what it blocks

Siberian box 55–75 N / 60–180 E, year 2099, 684 cells:

state	BNS	TREEFPC	GRASSFPC	grass > tree
offline CRUNCEP (the restart)	0.227	0.415	—	29.2 %
11E after 50 coupled years	0.017	0.083	—	69.6 %
PICAL_ccnice 2099	0.001	0.050	0.216	84.9 %

Stable between 2090 and 2099, so this is the settled state, and it is the end point of the transient round 27 identified: the coupled model dismantles an inherited forest, 92 % of the larch in 50 years. **The mechanism is**

competition with C3 grass, not a climate gate: BNS has no cold limit at all, the only temperature gate is a warm one 34 K away, and GDD5 clears on 91 % of cells in the very run where the larch dies. Questions about the boreal forest should be answered through the grass-versus-larch competition, not through thresholds.

This is now a practical blocker. Laszlo Hajdu’s LPJ-GUESS canopy work (stem area index via `ltos`, shrubs routed to low IFS vegetation, `iftreefracca 1`) measured +0.46 K global T2m in a run that *has* a boreal forest, and improved its ERA5 comparison (bias −0.46 against −0.92 K; RMSD 1.38 against 1.77). It acts on the tree side of the tile. With 5 % tree cover here, that result is not transferable, and a continuation would measure what the change does to a grassland. **PICAL_crunveg** — identical except that LPJ-GUESS starts from the CRUNCEP spin-up end state — was built to answer this and cancelled before starting when the campaign moved to albedo.

11 Model skill: the degradation is tropical

CMPI, lower is better, across four windows of the same run:

	1980–89	2000–09	2080–89	2090–99
CMPI global	0.844	0.879	0.902	0.903
nino34	0.966	1.064	1.124	1.121
tropics	0.877	0.874	1.013	1.011
northmid	0.823	0.834	0.867	0.869
arctic	0.867	0.888	0.880	0.871
antarctic	0.718	0.790	0.717	0.726

The polar regions are flat (+0.003 arctic, +0.008 antarctic across 110 years). The tropics carry the whole degradation (+0.155 nino34, +0.134 tropics). By variable the worst movers are **tas** +0.149, **thetao** +0.127, **zg** +0.111. **No mechanism is attached to this yet**, and it is not where the campaign has been looking. For scale, the campaign’s best-scoring run was 11R at 0.7577.

12 Levers, across the whole campaign

lever	verdict	the number
G4 (RVRSMIN cap)	adopted	Siberian JJA +0.952 K, $t = 7.68$; SO SW RMSE 6.88 → 4.80.
K1 (snow-free albedo)	adopted	+1.036 K, RMSE 1.579 → 1.524, no Arctic penalty.
P3/P5 (ECE_SNOW_SCF 3)	adopted	Refitted to snow courses; beats scheme-off on bias, RMSE and JJA against 105 RIHMI stations.
RSBLB 2.0	adopted	Arctic DJF screen +0.614 K coupled, net TOA null. The first lever to move winter and summer the same way.
GM 2500 / Redi 1000	adopted	Lowest TOA plateau kept, +0.70 W m ^{−2} . Costs NH ice, AMOC, and 16.7 Sv of Drake.
cvmix_TKE+IDEMIX	adopted	Closest to the observed Antarctic summer pack; costs NH April volume.
cavity_gamma_scale 0.6	adopted	1190 Gt yr ^{−1} ; upstream, FESOM #1064.
S4 RCL_INPPMIN 50000	retired	Headline of the 2026–08–11 summary. Siberian JJA +1.194 K AMIP, −0.365 K coupled.
I3	rejected	Strongest summer lever built (+1.322 K) at +1.72 W m ^{−2} at 60–90 N.
tanh snow depletion	falsified	−20.2 K against 174 676 station soil observations; unrepresentable complete cover.
LAMSK, moss proxy, D1	falsified	Skin conductivity +0.125 K against a 0.41 threshold; the moss proxy moved monotonically the wrong way; RCAPDCYCL 4 null.
Raupach z_0	dead structurally	Needs a canopy; makes the winter surface 4.4× smoother as the model stands.
DMS / marine CCN	priced out	Best SO lever found (−7.25 W m ^{−2} DJF) but the tropical tolerance binds below the published coefficient floor.
RSNOWLIN2 0.04	not adopted	Only lever that moves the tropical longwave (+2.74); costs +0.42 globally.
SB3 mixing length	falsified	3.43 tropical SW per K against RSBLB ’s 0.88.

lever	verdict	the number
salt plumes, MLE, h0min ↓, all three h0min 0.5 → 1.0	excluded reverted	Round 37, no SH winter response in 20 years. Flux response as designed (1.71→1.10), state did not follow; 60–90S bias +0.82 → +1.45 K.
darkening albpnd SSS restoring	falsified excluded by policy	Wrong direction, and double-counts the ponds. Permanently.

13 Reversals on the record

Kept because a summary that only lists what survived misrepresents how the campaign learned.

- ~~S4 is the best configuration the campaign has.~~ Its benefit reverses sign coupled. The 2026–08–11 summary led with it.
- ~~The period offset is ≈ 1.12 K, so the boreal bias is ≈ -1.0 K.~~ Measured directly at +0.42 K; the bias is ≈ -2.0 . The method error was applying an *observed* epoch change to a model that does not reproduce it.
- ~~There is a -1.0 W m^{-2} structural leak in the coupled column.~~ It was the omitted snow-enthalpy term.
- ~~The boreal problem is competition, not climate.~~ Both, and the competition is driven by the forcing: warming recovers 59 % of the gap at +2 K while **grass>tree** stays at 43–45 %.
- ~~Cloud amount is unreachable, so the SO needs an opacity lever.~~ Round 33 showed opacity is the right correction for 90–60 S and the *wrong* one for 60–45 S, which every campaign lever had been applying.
- ~~The ice is too dark.~~ It is too diffuse: compact-ice albedo is 0.760–0.778, but no cell exceeds 90 % concentration in January–March.
- ~~Positive net TOA means the run is warming.~~ It cools at 0.21 K/decade.

14 Bugs and traps that cost this campaign

what	the lesson
A dead namelist block	NCMIPFIXYR sat in &NAMECECMIP6, which the runtime does not read, so the AMIP runs were never at 1850 forcing. Krakatoa in 1883–86 is the fingerprint.
Two OASIS traps	oasis_put returns the waitgroup code for every field in a group but the last, so the ice-skin anchor was 0.0 in all 63 923 logged events and the skin reached -36.470 K. And GSMART on a sign-changing field gave 9105 excursions above 1 kW m^{-2} onto real ice in one coupled year, where GSSMAR and GLOBAL give none.
Implicit netCDF fill value	read_other_NetCDF left miss at 0.0 when a variable declared no missing_value , masking $146\,694$ real zeros and admitting 9.96921×10^{36} . FESOM PR #1087.
MPI_DOUBLE_PRECISION on a real(WP) buffer	Five MPI_AllREDUCE calls read 8 bytes from a 4-byte variable in a single-precision build. Not only a diagnostic: the corrupt total was fed back into the physics through idemix_botforc_Etot . Same PR.
The staged library is a symlink	<exp>/bin/fesom/lib/fesom/libfesom.so points into esm_tools ’ per-leg snapshots and md5sum follows it. The fesom executable is a driver stub, byte-identical across builds, so a matching binary checksum proves nothing. Use ls -l before md5sum . Cost a 47-node slot and a NaN’d ocean.
The instruction file nothing reads	guess.ins is hard-wired in framework.cpp:1063 and imports ecearth.ins ; global.ins and run_coupled_4_1_2.ins are staged and never reached. The two had drifted apart, ifreefracca 1 against 0. A parameter put in the wrong one is silently inert.
Chained legs and the date file	Legs read a per-leg archive, not the source runscript, so edits need -U ; and the date file advances past a crashed leg, so it is not evidence of progress.

15 Closed by measurement

Claims that were live at some point and are now settled, each with the observation that closed it.

claim	what closed it
The cold bias is a tuning artefact	Centred RMSD is 1.41–1.64 K across all 29 early runs, a $\pm 8\%$ spread. The runs differ by an offset.
Land surface albedo explains the universal bias	Removing 74 % of a measured bias bought 0.089 K of ≈ 0.7 K.
Boreal needleleaf is excluded by a cold gate	True for BNE (0.7 % of cells clear it) and false for BNS, which is $\approx 80\%$ of NE Siberia and is outcompeted instead.
ERA5 is an adequate cloud reference for the Southern Ocean	ERA5 is IFS and reproduces the model's own bias; CERES and MODIS agree with each other against both.
The tropical deep warming is diapycnal flux	It is heave: 0.42 of the +0.477 K at fixed depth is the isopycnal sinking +42.4 m.
The SH winter bias is a vertical-mixing deficiency	Four levers, four nulls, 20 years each from one restart.
The ice is too dark because the ponds are too bright	The <code>ice_therm</code> albedos are pond-free; darkening them double-counts.
The albedo sensitivities are fixed properties	NH $\partial\alpha/\partial\text{albsn}$ moved $0.149 \rightarrow 0.504$ as the Arctic cooled.
Positive net TOA means the run is warming	It cools at 0.21 K/decade; the ocean redistributes rather than accumulates.
GM tuning is free at the Southern Ocean	16.7 Sv of Drake Passage transport.

16 Open, in priority order

1. **The upper-ocean heat drain.** 0.32 W m^{-2} equivalent out of the top 700 m is the mechanism behind the cold drift and nothing has been aimed at it. Mixing scheme, Southern Ocean freshwater regime, or both.
2. **NH April volume** decides whether TKE+IDEMIX keeps its place: $+1.145 \pm 0.968/\text{decade}$, sitting +3.37 above the early GIOMAS record.
3. **The Antarctic at 62 % of observed winter volume**, unmoved by any Round 37 lever and not a vertical-mixing problem.
4. **The ACC 30 % weak and slowly weakening**, with the GM setting that improved the mid-depth uptake making it worse. Unreconciled.
5. **The Southern Ocean cloud amount deficit**, 84 % of it inherited from AMIP and unmoved by any lever in the campaign.
6. **Does the canopy work do anything without trees?** PICAL_crunveg was built to answer it and cancelled before starting.
7. **Will the CRUNCEP state load on TC095-land?** Written on an 11538-cell L096 gridlist; a missing-cell case was once fatal with peatland on and the code has changed substantially since. Untested.
8. **The tropical CMPI degradation** has no mechanism attached.
9. **The residual deep land cold bias**, $\approx 1.2 \text{ K}$ still present at 850 hPa, out of reach of any mixing lever.

17 What would change the picture

Stated in advance, so that a result which contradicts them is recognised as such.

- **If the upper-ocean heat drain stops without the surface warming**, then the cold drift is not driven by the redistribution and the diagnosis in §5 is wrong.
- **If NH April volume flattens on its own** over the next 50 years, the cost priced against TKE+IDEMIX at adoption was an artefact of the 20-year test arm and the scheme is cheaper than believed.
- **If a Southern Ocean lever moves the winter bias in under 20 years**, then the two-regime picture is wrong, or the regime boundary is closer than the five excluded levers suggest.
- **If Drake Passage recovers without GM being reduced**, the 16.7 Sv attribution is confounded with something else that changed between PI200 and the arms.
- **If the canopy work produces +0.46 K here**, with 5 % tree cover, then it is not acting through the tree canopy and the mechanism is misattributed.
- **If the tropical CMPI degradation appears in a run without the cold drift**, the two are independent and the tropics need their own investigation.

- If closing the remaining cold bias does not recover the forest, the round-27 transfer function does not transfer coupled, and the campaign’s founding premise fails at the last step.

18 The run inventory since the last summary

Every coupled arm that produced a scoreable result since 2026–08–24. All are 20 years unless stated, all branch from one restart with the control on identical years. The campaign’s earlier arms — the 29 screening runs, the AMIP A/D/F/G/I/K/L/P/S series and the 11-series coupled roster — are catalogued in `report.tex`, appendix `app:params`.

run	years	the one thing it changed, and what it showed
PI200_spp / _mle / _h0 / _all3	1560–79	Salt plumes, Fox-Kemper MLE, lead closing <code>h0min</code> 0.5 → 0.25 m, and all three together. No SH winter response in any, and no combination effect.
PI200_cav05/06/07	1600	One-year basal-melt screens: 1038 / 1190 / 1402 Gt yr ⁻¹ against 1893 at 1.0. 0.6 taken forward.
PI200_cavg06	1600–19	<code>cavity_gamma_scale</code> 0.6. Melt calibrated; no Southern Ocean response.
PI200_gmR1500	1580–99	Rossby cutoff, K_{GM}^{max} 1500. Drake 103.2 Sv.
PI200_gmR2500	1580–99	Rossby cutoff, K_{GM}^{max} 2500. Drake 93.3 Sv; the adopted setting, and the 16.7 Sv cost.
PICAL_v35def	2020–49	<code>RCL_INPPMIN</code> 50000 vs 70000, paired with momix-off. Context for the TOA figure.
PICAL_v35def_darkice	2050–99	<code>albi/albim</code> darkened against its parent. An Arctic runaway fifty years did not arrest.
PICAL_cvKPP	2020–39	<code>cvmix_KPP</code> . The useful null: reproduces the KPP control.
PICAL_cvTKE	2020–39	<code>cvmix_TKE</code> . Grows the Antarctic summer pack; costs energy.
PICAL_cvTKEIDEMIX	2020–39	<code>cvmix_TKE+cvmix_IDEMIX</code> . Closest to the observed summer pack; adopted.
PICAL_ccnice	1940–2099	The production line.
PICAL_crunveg	—	Built to test the canopy work on a forested land state; cancelled before starting.

19 The evaluation protocol, and which references are independent

Every window is scored the same way, so results from different rounds are comparable.

target	reference, and its caveat
Sea-ice volume	GIOMAS . A model with assimilation, not a measurement: Arctic volume carries 10–20 % uncertainty and Antarctic more. Use 1989–99 for a PI run — the Arctic thinned sharply across the full record, the Antarctic has no trend.
Sea-ice extent	OSI-SAF . The record on disk has no December — 286 months — and a December value must not be synthesised.
Radiation	CERES . Period offsets matter, and measuring one rather than estimating it once reversed a conclusion.
Screen temperature	ERA5 , except for anything the IFS land surface decides: ERA5 is an HTESSEL sibling and reproduces the model’s own cloud and snow biases.
Soil temperature	RIHMI-WDC station data, 174676 observations. This is why the campaign no longer depends on ERA5 for a soil claim.
Snow cover	Rutgers and Russian snow courses (14369 Siberian DJF surveys).
Cloud amount	CERES and MODIS , which agree with each other, over dark ocean only — over ice they disagree by 7.7 pp.
Ocean T/S	PHC3 on the <i>new</i> CORE3 mesh (220509 nodes). The 211567-node beta mesh climatology is a different file and mixing them fails silently.
Drake Passage	Cunningham 2003 134 ± 11.2 Sv and Donohue 2016 173.3 ± 10.7 Sv. Donohue includes near-bottom flow a 47-level mesh cannot carry, so the gap to 134 is the real bias.
Cavity basal melt	Rignot 2013 ~ 1500 and Adusumilli 2020 ~ 1100 Gt yr ⁻¹ ; the calibration target is 1100–1325.
Composite skill	cmpitool against ERA5, thirteen variables by region and season. Useful for regressions, blunt for mechanisms.

How long an arm has to run. The standing threshold is $2\sigma\sqrt{2/N}$ from the control’s own scatter in the same window, with no autocorrelation correction, so it is a floor rather than a true confidence interval — and branched arms are scored *paired*, because they share a restart and the unpaired form discards that and once hid a resolved result. Measured on this configuration: Drake Passage scatter ~ 1.5 Sv; ice volume needs 20 years to resolve $\pm 1 \times 10^3$ km³/decade; decadal TOA means are stable to $\sim \pm 0.1$ W m⁻²; wallclock varies $\pm 15\%$, so SYPD comparisons need more than one leg each. The per-band one-year thresholds from round 26 remain in use: SO SW CRE annual ± 1.97 , SO DJF ± 3.82 , tropics annual ± 0.99 , global ± 1.11 .

The full per-window diagnostic set is `reval` (`release_evaluation_tool2`), driven by one config per window. Copying `configs/awiesm3_v35_pical_ccnice_2090s.py` and changing four fields (`model_version`, `pi_ctrl_start/end`, `spinup_end`, `out_path`) is the whole procedure; it submits 26 jobs and an HTML report.

20 Method notes

Techniques that earned their keep, recorded because they generalise.

Measure the reference-period offset, do not estimate it. Applying an *observed* epoch change to a model that does not reproduce that change put the boreal bias off by a factor of two and cost a withdrawn conclusion. The AMIP present-day run measures it directly.

Verify the artefact that executes, not the one that was edited. A namelist parameter is live only if the *staged* shared library carries it, and the file that is read may not be the one that looks live — LPJ-GUESS reads a hard-wired `guess.ins` while several plausible instruction files are staged beside it and never opened. Both traps cost a run each.

Prove a physical diagnostic against the input file’s own metadata. The IDEMIX gate works because 0.9179869 TW is the forcing file’s `Etot_in_TW_bin` attribute, not a remembered number. A self-consistency check would have passed on the broken build.

Use the model’s own integrals where they exist. Hemispheric ice volume comes from FESOM’s `sivoln/sivol`s, computed on the native mesh inside the model, so it cannot pick up the wrong mesh. Likewise Drake Passage is computed on the model’s own transect path rather than by interpolating to a straight line, which would not conserve.

Screen at one year, or offline, before spending forty. The basal-melt calibration was settled by three one-year runs; the snow scheme was validated offline before implementation; the forcing-transfer test used two spin-ups that already existed.

Keep one store, append to it. `data/coupled_annual_diag.nc` holds annual TOA, surface and four-band ocean heat content for 90 runs across 750 model years, and survives the deletion of the raw output it was built from. Several results here would be unrecoverable otherwise.

21 The software stack, and the move to albedo

component	branch	state
esm_tools	feat/awiesm3-v3.4-co2	cf0a27943. Carries the v3.5 CORE3 defaults, the mixing scheme and the IDEMIX forcing paths.
oifs-48r1	awiesm3-develop	At its remote tip.
FESOM	main + PR #1060 + PR #1087	Both PRs still open. What <code>esm_master</code> installs (<code>feat/cavity-gamma-scale</code>) is 53 commits behind <code>main</code> and lacks both IDEMIX fixes, so it cannot run this experiment.
LPJ-GUESS	lpj_guess_awiesm3 @ 1325440	The AWI land-grid line merged with the canopy work, 2026-09-25. Built as <code>guess.4112_awiesm3canopy_1325440_20260925</code> .
XIOS, rnfmmap, OASIS	main, enthalpy_on_ocean_side, local_combined_fixes	Fine as installed.

Merged upstream during the campaign: FESOM PR #1054 (the single-precision ocean heat leak), #1064 (`cavity_gamma_scale`), and three CI fixes of our own that unblocked stacked pull requests in that repository (#1088, #1089, #1091). Still open and needed: #1060 (ice skin from the atmosphere) and #1087 (the two CVMix bugs).

The campaign is moving to albedo for queue time: a 47-node leg on levante has waited between 30 seconds and 10 hours to start. The branch point is 2100, the only year with a complete consistent restart set. Three

one-liners kill a first run there, all in `HANDOFF_albedo_spinup_continuation.md`: `h0min` in the runscript aborts FESOM at the namelist read on any branch except the abandoned `feat/leadclose-h0min`; both IDEMIX forcing files must exist under `${fesom.pool_dir}/forcing/idefix/`; and `ltos` must go in `ecearth.ins.j2`, the file the run actually reads. The standing gate on any IDEMIX leg is `Etot_srf` 0.3211645 and `Etot_bot` 0.9135067 $\rightarrow$ 0.9179869 TW within the first minute of the compute log; the failure signature is 6.05×10^{-4} with -1.54×10^{16} , which NaN'd the ocean in 100 seconds on 2026-09-24.

22 The path forward

In the order that would resolve the most with the least compute.

- 1. Attack the upper-ocean heat drain.** It is the mechanism behind the cold drift, it is quantified (0.32 W m^{-2} equivalent), and nothing has been aimed at it. The first question is whether it is the mixing scheme or the Southern Ocean freshwater regime, and that is answerable by comparing the drain between the four mixing arms, which already exist. **No new compute required for the first step.**
- 2. Settle whether the canopy work does anything here.** `PICAL_crunveg` is built and costs 50 years. It also answers, as a by-product, whether the CRUNCEP state loads on `TC095-land` at all, which gates every future land experiment on this configuration.
- 3. Reconcile the ACC against the mid-depth uptake.** GM 2500 improved one and cost 16.7 Sv of the other. The cheap version is to score Drake Passage on the arms that already exist before proposing a new GM setting, so the next choice answers to both constraints.
- 4. Give the tropical CMPI degradation a mechanism.** The largest unexplained skill loss in the run, and the campaign has never looked there. It may be a symptom of the upper-ocean drain rather than an independent problem.
- 5. Decide what the Arctic is for.** NH April volume is +3.37 above the early GIOMAS record and rising. Either that is a bias to tune out, or the run is being asked to match a satellite-era record it should legitimately exceed — and until that is fixed, the cost side of every ice-affecting lever is priced against a target nobody has agreed.

Note what is *not* on this list. The atmospheric lever families are, by measurement, nearly exhausted: land albedo is closed, surface exchange is retired as a class, the snow scheme is refitted, cloud amount is unreachable and the one tropical longwave lever costs more globally than it buys. The remaining leverage is in the ocean and in the vegetation, which is where both open items sit.

23 Assets

what	where
Full record, 148 pages	<code>report/report.tex</code>
Previous summary (2026-08-24)	<code>report/summary_superseded_20260926.tex</code>
Albedo handoff	<code>HANDOFF_albedo_spinup_continuation.md</code>
Annual TOA / surface / OHC store	<code>data/coupled_annual_diag.nc</code> , built by <code>scripts/analysis/coupled_annual_store.py</code>
Drift, ice volume, GIOMAS	<code>scripts/analysis/ccnice_regional_drift.py</code> , <code>ice_volume_series.py</code> , <code>giomas_volume_targets.py</code> (takes <code>OBS_Y0/OBS_Y1</code>)
Drake Passage, cavity melt	<code>scripts/analysis/drake_passage_transport.py</code> and <code>scripts/figures/cavity_melt_maps_mixing.py</code> ; both need the conda env <code>esm-tools_auto_tripyview</code> , and the transport needs <code>PYTHONPATH=/work/ab0246/a270092/software/tripyview</code>
Run and parameter notes	<code>notes/RUNS_AND_PARAMETERS.md</code>
Full reval diagnostics	<code>/work/ab0246/a270092/software/release_evaluation_tool2</code> , per-window configs under <code>configs/</code>
Configuration	<code>esm_tools feat/awiesm3-v3.4-co2</code> at <code>cf0a27943</code>